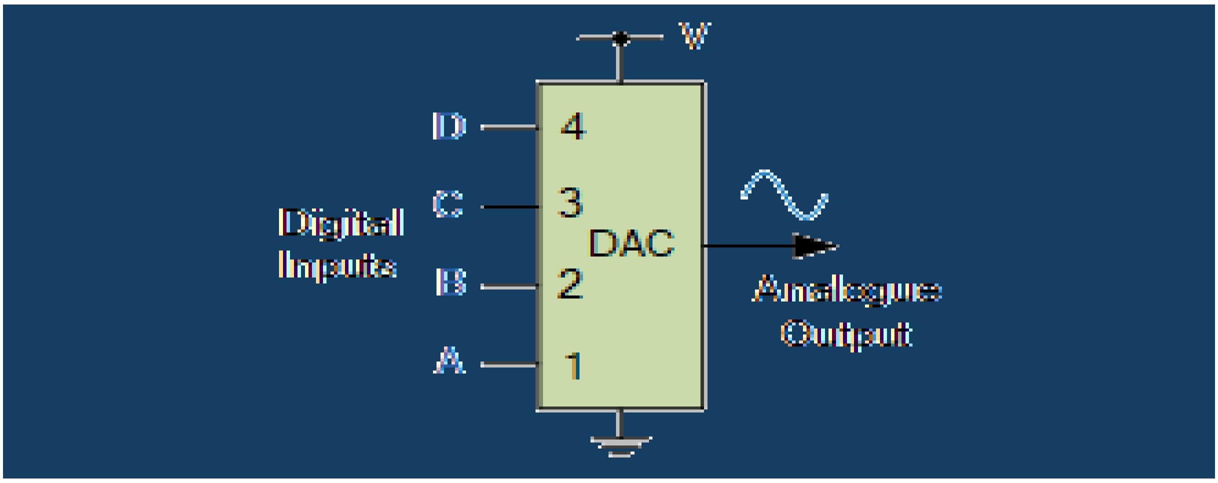


Binary Weighted DAC

Binary weighted digital-to-analogue converters are a type of data converter which converts a digital binary number into an equivalent analogue output signal proportional to the value of the digital number

The **Digital to Analogue Converter**, or **DAC's** as they are more commonly known, are the opposite of the *Analogue to Digital Converter* we looked at in a previous tutorial. DAC's convert binary or non-binary numbers and codes into analogue ones with its output voltage (or current) being proportional to the value of its digital input number.

For example, we may have a 4-bit digital logic circuit that ranges from 0000 to 1111_2, (0 to F_{16}) which a DAC converts to a voltage output ranging from 0 to 10V.

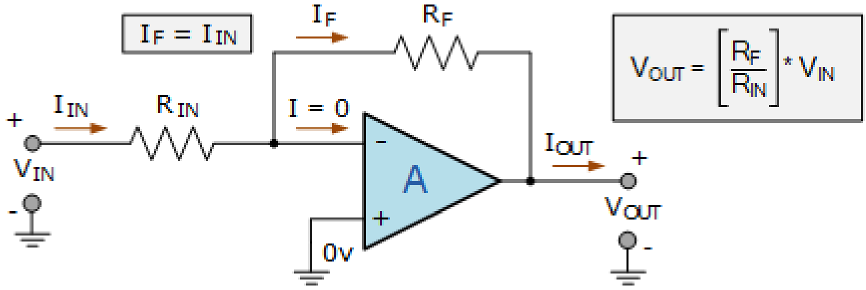
Converting an "n"-bit digital input code into an equivalent analogue output voltage between 0 and some V_{MAX} value can be done in a number of ways, but the most common and easily understood conversion methods uses a weighted resistors and a summing amplifier, or a R-2R resistor ladder network and operational amplifier.

Both *digital-to-analogue conversion* methods produce a weighted sum output, with the weights set by the resistive values used in the ladder networks contributing a different "weighted" amount to the signals output.

Operational Amplifiers such as the inverting op-amp circuit uses negative feedback to both reduce and control its very high open-loop gain, A_{OL}. It does this by feeding back a small fraction of its output signal back to its input terminal.

For an inverting amplifier, the input voltage V_{IN} is connected directly to its inverting input via a suitable input resistor R_{IN} and that the inverting amplifiers closed-loop voltage gain, $A_{V(CL)}$ is determined by the ratio of these two resistors as shown.

Inverting Operational Amplifier Circuit



Then we can see that V_{OUT} is given as V_{IN} multiplied by the closed-loop Gain (A_{CL}), which is determined by the ratio of the feedback resistance, R_F to the input resistance, R_{IN}. So by altering the values of either R_F or R_{IN} we can change the closed-loop gain of the op-amp and therefore the value of V_{OUT} (I_F*R_f) for a given input signal.

Here in this inverting operational amplifier example we have used a single input voltage signal, but what if we added another input resistor to combine two or more analogue signals into a single output, what would be the effect on the circuit and its gain.

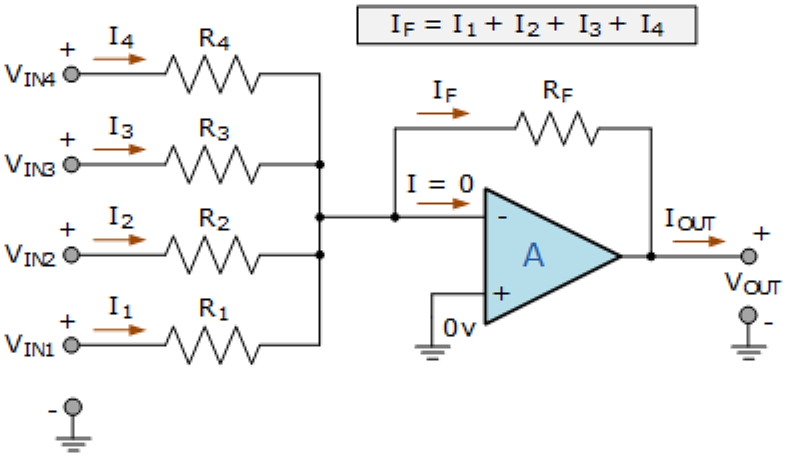
Digital-to-Analogue Converter Summing Amplifier

By connecting multiple inputs to the negative terminal of the operational amplifier, we can convert the single input circuit from above into a *summing amplifier* or to be more precise, a “summing inverting voltage amplifier” circuit.

As the negative feedback created by the feedback resistor, R_F biases the inverting input of the op-amp at zero potential, any input signals are effectively electrically isolated from each other with the output being the inverted sum of all the input signals combined.

Thus a summing amplifier in the inverting mode produces the negative sum of any number of input voltages, whereas a no-inverting summing amplifier would produce the positive sum of any number of input voltages. Consider the circuit below.

Inverting Summing Amplifier Circuit



In the summing amplifier circuit above, the output voltage, (V_{OUT}) is proportional to the sum of the four input voltages, V_{IN1}, V_{IN2}, V_{IN3}, and V_{IN4} and we can modify the original equation for the inverting amplifier configuration above to take account of these four new input values as follows:

$$I_F = I_1 + I_2 + I_3 + I_4 = \frac{V_{IN1}}{R_1} + \frac{V_{IN2}}{R_2} + \frac{V_{IN3}}{R_3} + \frac{V_{IN4}}{R_4}$$

$$V_{OUT} = \frac{R_F}{I_F} \times V_{IN} = -\left(\frac{R_F}{R_1} V_{IN1} + \frac{R_F}{R_2} V_{IN2} + \frac{R_F}{R_3} V_{IN3} + \frac{R_F}{R_4} V_{IN4} \right)$$

$$\therefore V_{OUT} = -\frac{R_F}{R_{IN}} \left(V_{IN1} + V_{IN2} + V_{IN3} + V_{IN4} \right)$$

Then we can see that the output voltage is an inverted, scaled sum of the four input voltages as each input voltage is multiplied by its corresponding gain and added to the next to produce the total output.

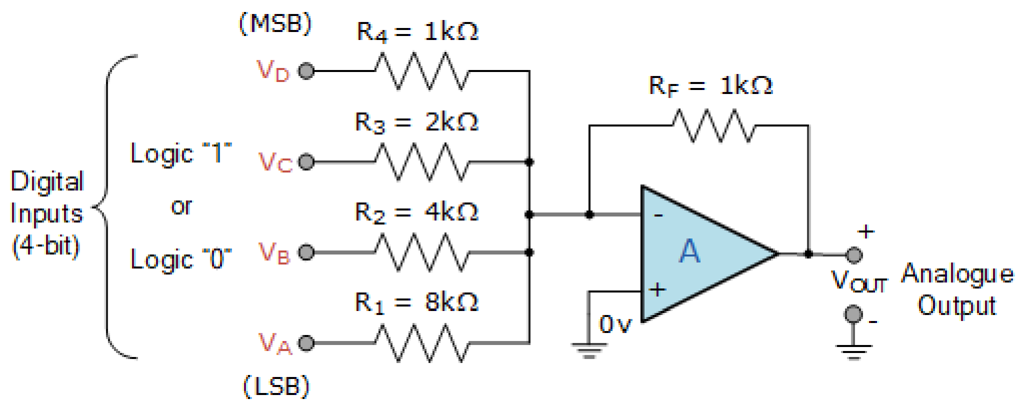
If all the resistances are the same and of an equal value, that is: $R_F = R_1 = R_2 = R_3 = R_4$, then each input channel will have a closed-loop voltage gain of unity (1) so the output voltage is given simply by:

$$V_{OUT} = -(V_{IN1} + V_{IN2} + V_{IN3} + V_{IN4})$$

So if we now assume that the four inputs of the summing amplifier are binary inputs with voltage values of either 0 or 5 volts (LOW or HIGH, 0 or 1) and we double the resistive values of each input resistor with regards to the previous one, we can produce an output condition which would be the weighted sum of these four input voltages creating the basic circuit for a 4-bit binary weighted digital-to-analogue converter, or 4-bit weighted D/A converter.

Labelling the four summing inputs as A, B, C, D and making R_F = 1kΩ, with the four input resistors ranging from 1kΩ to 8kΩ (or multiples thereof), we can construct a simple 4-bit binary weighted analogue-to-digital converter circuit as shown.

4-bit Binary Weighted Digital-to-Analogue Converter



For a 4-bit binary number there are 2^4 = 16 possible combinations or A, B, C, and D ranging from 0000_2 to 1111_2 which corresponds to decimal 0 to 15 respectively. If we make the weight of each input bit double with respect to the other, we end up with an 8-4-2-1 binary code ratio corresponding to 2^3, 2^2, 2^1 and 2^0.

So if we set the "D" input resistance at 1kΩ, the "C" input resistance at 2kΩ (that is the double of D), the "B" input resistance at 4kΩ (double C), and the "A" input resistance at 8kΩ (double B), with the feedback resistance R_F set again at 1kΩ, then the transfer characteristic of the 4-bit binary weighted digital-to-analogue converter would be:

4-bit DAC Transfer Characteristic

$$V_{OUT} = -\left[\frac{R_F}{R_4}V_D + \frac{R_F}{R_3}V_C + \frac{R_F}{R_2}V_B + \frac{R_F}{R_1}V_A\right]$$

$$V_{OUT} = -\left[\frac{1k\Omega}{1k\Omega}V_D + \frac{1k\Omega}{2k\Omega}V_C + \frac{1k\Omega}{4k\Omega}V_B + \frac{1k\Omega}{8k\Omega}V_A\right]$$

$$V_{OUT} = -\left[1V_D + \frac{1}{2}V_C + \frac{1}{4}V_B + \frac{1}{8}V_A\right]$$

So we can see that if a TTL voltage of +5 volts (logic 1) is applied to the summing amplifiers input, V_D which represents the most significant bit (MSB), the op-amp's gain will be R_F/R_4 = 1kΩ/1kΩ = 1 (unity). Thus with a 4-bit binary code of 1000 applied, the output of the digital-to-analogue converter circuit will be -5 volts.

Likewise, if +5 volts (logic 1) is applied to the summing amplifiers input V_C, the op-amp's gain will be R_F/R_3 = 1kΩ/2kΩ = 1/2 (one half). So the 4-bit binary code of 0100 would produce an analogue output voltage of -2.5 volts.

Again with a logic "1" applied to the summing amplifiers input V_B, the op-amp's gain will be R_F/R_2 = 1kΩ/4kΩ = 1/4 (one quarter) with the 4-bit binary code of 0010 producing an output voltage of -1.25 volts.

Finally a logic "1" applied to the summing amplifiers input, V_A which represents the least significant bit (LSB), the op-amp's gain will therefore be R_F/R_1 = 1kΩ/8kΩ = 1/8 (one eighth) with the 4-bit binary code of 0001 producing an output voltage of -0.625 volts, (a 12.5% resolution).

The resolution of this simple 8-4-2-1 binary weighted digital-to-analogue converter will produce an output voltage change of 0.625 volts per 1-bit change in the binary number, and we can express this output voltage change in the following table.

4-bit Binary Weighted D/A Converter Output

Digital Inputs				V_{OUT} Expression	V_{OUT}
D	C	B	A	$1*V_D + {}^1/_2*V_C + {}^1/_4*V_B + {}^1/_8*V_A$	in Volts
0	0	0	0	0*5 + 0*5 + 0*5 + 0*5	0
0	0	0	1	0*5 + 0*5 + 0*5 + $^1/_8$*5	–0.625
0	0	1	0	0*5 + 0*5 + $^1/_4$*5 + 0*5	–1.25
0	0	1	1	0*5 + 0*5 + $^1/_4$*5 + $^1/_8$*5	–1.875
0	1	0	0	0*5 + $^1/_2$*5 + 0*5 + 0*5	–2.50
0	1	0	1	0*5 + $^1/_2$*5 + 0*5 + $^1/_8$*5	–3.125
0	1	1	0	0*5 + $^1/_2$*5 + $^1/_4$*5 + 0*5	–3.75
0	1	1	1	0*5 + $^1/_2$*5 + $^1/_4$*5 + $^1/_8$*5	–4.375
1	0	0	0	1*5 + 0*5 + 0*5 + 0*5	–5.00
1	0	0	1	1*5 + 0*5 + 0*5 + $^1/_8$*5	–5.625

1	0	1	0	1*5 + 0*5 + ¼*5 + 0*5	–6.25
1	0	1	1	1*5 + 0*5 + ¼*5 + ⅛*5	–6.875
1	1	0	0	1*5 + ½*5 + 0*5 + 0*5	–7.50
1	1	0	1	1*5 + ½*5 + 0*5 + ⅛*5	–8.125
1	1	1	0	1*5 + ½*5 + ¼*5 + 0*5	–8.75
1	1	1	1	1*5 + ½*5 + ¼*5 + ⅛*5	–9.375

Where the output voltages are all negative due to the inverting input of the summing amplifier.

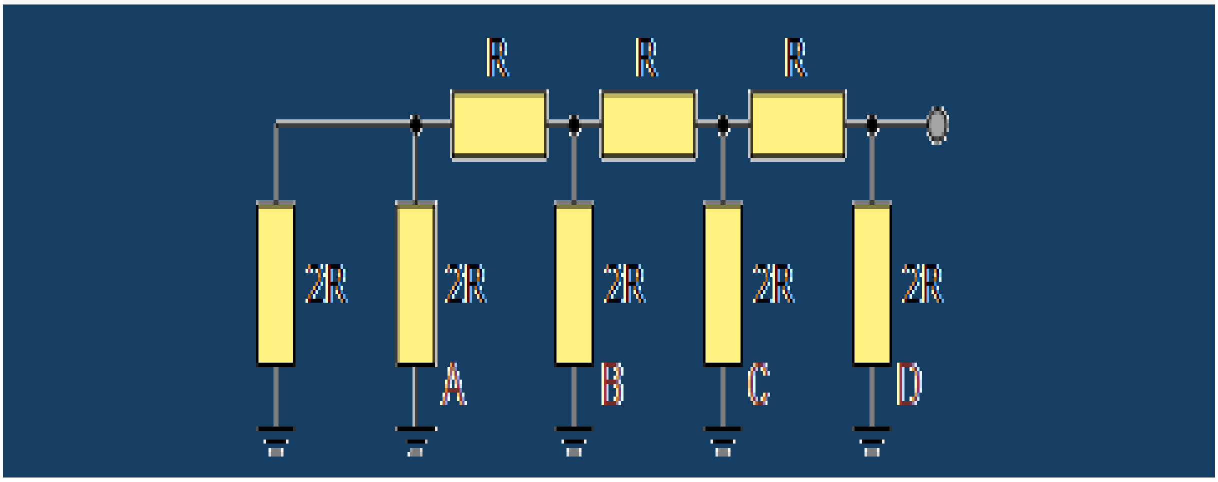
By increasing the number of binary digits and the resistive summing network so that each resistor has a different weighting, the resolution of the analogue output voltage for a binary weighted digital-to-analogue converter can be increased.

For example, an 8-bit DAC with TTL +5 inputs would produce a resolution of 0.039 (1/128*V) volts, while a 12-bit DAC would be 0.00244 (1/2048*V) volts per step (1 LSB) change of the input binary (or non-binary) code.

Clearly then the disadvantage here is that a binary weighted resistor DAC requires a large range of high precision resistors (one per bit) for an "n"-bit DAC making it impractical (and expensive) for converters with more than a just a few bits of resolution.

But we can expand on this idea of a binary weighted digital-to-analogue circuit configuration which uses different value resistors one step further by converting it into a R-2R resistor ladder DAC which requires only two precision resistance values, namely R and 2R.

R-2R DAC



R-2R DAC

R-2R Digital-to-Analogue Converter, or DAC, is a data converter which use two precision resistor to convert a digital binary number into an analogue output signal proportional to the value of the digital number

Compared to the R-2R DAC, the **binary weighted digital-to-analogue converter** has an analogue output voltage which is the weighted sum of the individual inputs. Thus it requires a large range of precision resistors within its ladder network, making its design both expensive and impractical for most DAC's requiring lower levels of resolution.

As the binary weighted DAC is based on a closed-loop inverting operational amplifier using summing amplifier topology, this type of data converter configuration may work well for a D/A converter of a few bits of resolution. But a much simpler approach is using a R-2R resistive ladder network to construct a **R-2R Digital-to-Analogue Converter** requires only two precision resistances.

The R-2R resistive ladder network uses just two resistive values. One resistor has the base value "R", and the second resistor has twice the value of the first resistor, "2R", no matter how many bits are used to make up the ladder network.

So for example, we could just use a standard 1kΩ resistor for the base resistor "R", and therefore a 2kΩ resistor for "2R" (or multiples thereof as the base value of R is not too critical). Thus the resistive value of 2R will always be twice the value of the base resistor, R. That is 2R = 2*R. This means that it is much easier for us to maintain the required accuracy of the resistors along the ladder network compared to the previous weighted resistor DAC. But what is a “R-2R resistive ladder network” anyway.

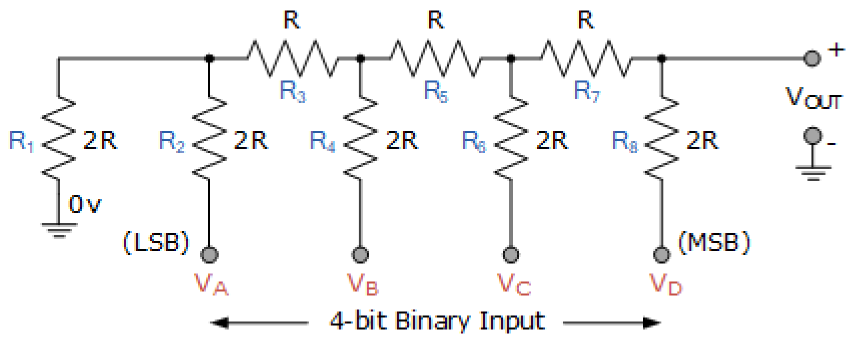
R-2R Resistive Ladder Network

As its name implies, the “ladder” description comes from the ladder-like configuration of the resistors used within the network. A R-2R resistive ladder network provides a simple means of converting digital voltage signals into an equivalent analogue output.

Input voltages are applied to the ladder network at various points along its length and the more input points the better the resolution of the R-2R ladder. The output signal as a result of all these input voltage points is taken from the end of the ladder which is used to drive the inverting input of an operational amplifier.

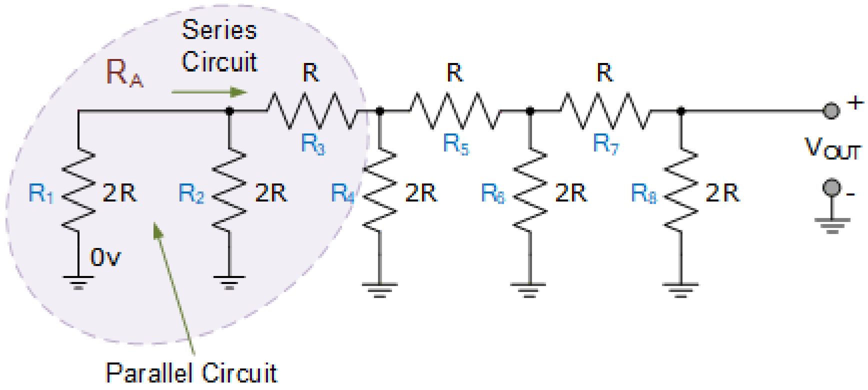
Then a R-2R resistive ladder network is nothing more than long strings of parallel and series connected resistors acting as interconnected voltage dividers along its length, and whose output voltage depends soley on the interaction of the input voltages with each other. Consider the basic 4-bit R-2R ladder network (4-bits because it has four input points) below.

4-bit R-2R Resistive Ladder Network



This 4-bit resistive ladder circuit may look complicated, but its all about connecting resistors together in parallel and series combinations and working back to the input source using simple circuit laws to find the proportional value of the output. Lets assume all the binary inputs are grounded at 0 volts, that is: $V_A = V_B = V_C = V_D = 0V$ (LOW). The binary code corresponding to these four inputs will therefore be: **0000**.

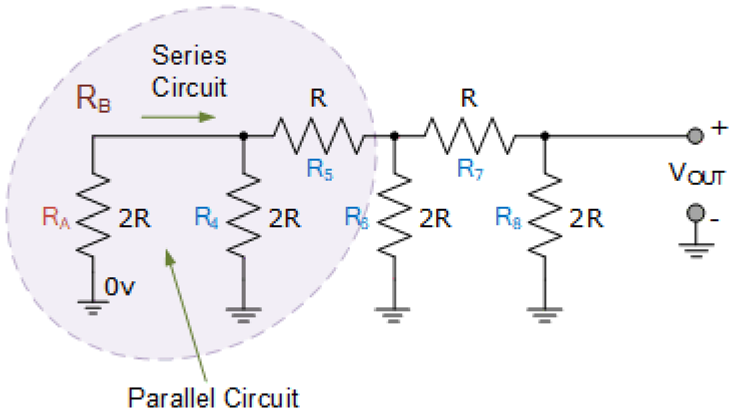
Starting from the left hand side and using the simplified equation for two parallel resistors and series resistors, we can find the equivalent resistance of the ladder network as:



Resistors R_1 and R_2 are in "parallel" with each other but in "series" with resistor R_3. Then we can find the equivalent resistance of these three resistors and call it R_A for simplicity (or any other form of identification you want).

$$R_A = R_3 + \frac{R_1 \times R_2}{R_1 + R_2} = R + \frac{2R \times 2R}{2R + 2R} = R + R = 2R$$

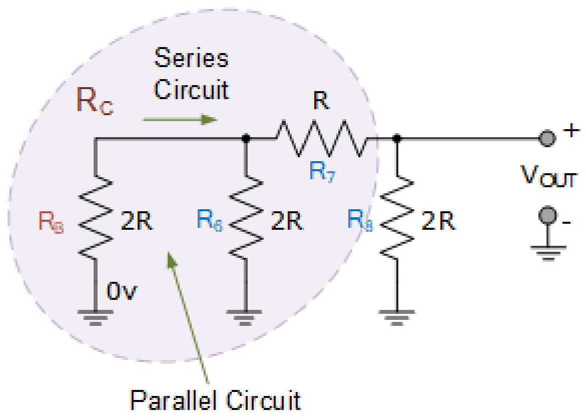
Then R_A is equivalent to "2R". Now we can see that the equivalent resistance "R_A" is in parallel with R_4 with the parallel combination in series with R_5.



Again we can find the equivalent resistance of this combination and call it R_B.

$$R_B = R_5 + \frac{R_A \times R_4}{R_A + R_4} = R + \frac{2R \times 2R}{2R + 2R} = R + R = 2R$$

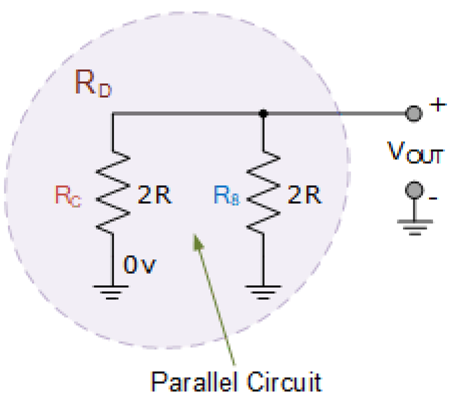
So R_B combination is equivalent to "2R". Hopefully we can see that this equivalent resistance R_B is in parallel with R_6 with the parallel combination in series with R_7 as shown.



As before we find the equivalent resistance and call it R_C.

$$R_C = R_7 + \frac{R_B \times R_6}{R_B + R_6} = R + \frac{2R \times 2R}{2R + 2R} = R + R = 2R$$

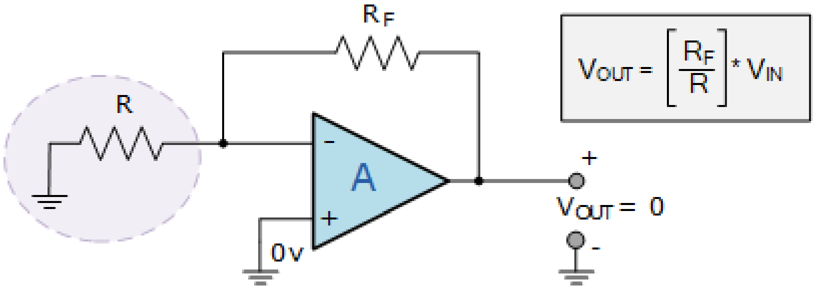
Again, resistor combination R_C is equivalent to "2R" which is in parallel with R_8 as shown.



As we have shown above, when two equal resistor values are paralled together, the resulting value is one-half, so 2R in parallel with 2R equals an equivalent resistance of R. So the whole 4-bit R-2R resistive ladder network comprising of individual resistors connected together in parallel and series combinations has an equivalent resistance (R_{EQ}) of "R" when a binary code of "0000" is applied to its four inputs.

Therefore with a binary code of "0000" applied as inputs, our basic 4-bit R-2R digital-to-analogue converter circuit would look something like this:

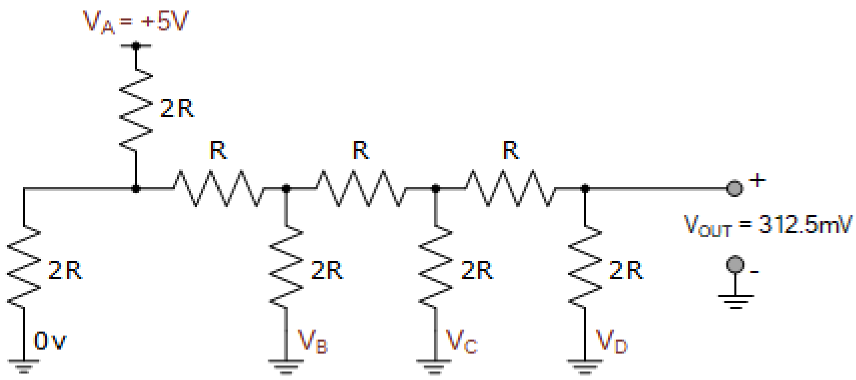
R-2R DAC Circuit with Four Zero (LOW) Inputs



The output voltage for an inverting operational amplifier is given as: $(R_F/R_{IN})*V_{IN}$. If we make R_F equal to R, that is R_F = R = 1, and as R is terminated to ground (0V), then there is no V_{IN} voltage value, (V_{IN} = 0) so the output voltage would be: (1/1)*0 = 0 volts. So for a 4-bit R-2R DAC with four grounded inputs (LOW), the output voltage will be "zero" volts, thus a 4-bit digital input of **0000** produces an analogue output of 0 volts.

So what happens now if we connect input bit V_A HIGH to +5 volts. What would be the equivalent resistive value of the R-2R ladder network and the output voltage from the op-amp.

R-2R DAC with Input V_A

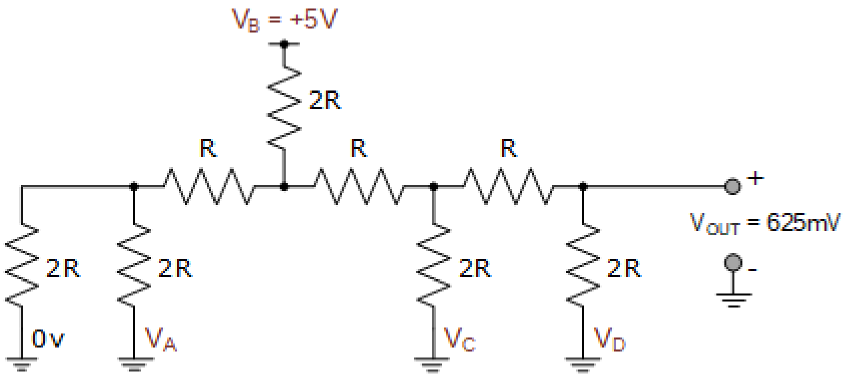


Input V_A is HIGH and logic level "1" and all the other inputs grounded at logic level "0". As the R/2R ladder network is a linear circuit we can find Thevenin's equivalent resistance using the same parallel and series resistance calculations as above to calculate the expected output voltage. The output voltage, V_{OUT} is therefore calculated at 312.5 milli-volts (312.5 mV).

As we have a 4-bit R-2R resistive ladder network, this 312.5 mV voltage change is one-sixteenth the value of the +5V input (5/0.3125 = 16) voltage so is classed as the Least Significant Bit, (LSB). Being the least significant bit, input V_A will therefore determine the "resolution" of our simple 4-bit digital-to-analogue converter, as the smallest voltage change in the analogue output corresponds to a single step change of the digital inputs. Thus for our 4-bit DAC this will be 312.5mV (1/16th) for a +5V input.

Now lets see what happens to the output voltage if we connect input bit V_B HIGH to +5 volts.

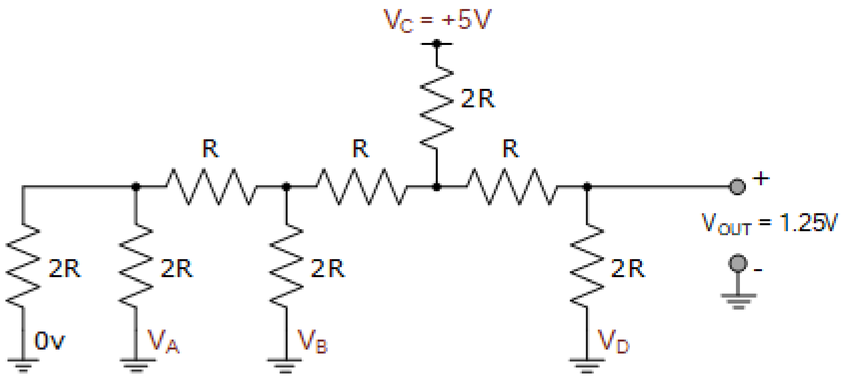
R-2R DAC with Input V_B



With input V_B HIGH and logic level "1" and all the other inputs grounded at logic level "0", the output voltage, V_{OUT} is calculated at 625mV, and which is one-eighth (1/8th) the value of the +5V input (5/0.625 = 8) voltage. We can also see that it is double the output voltage when only input bit V_A was applied, and we would expect this as its the 2_{nd} bit (input) so has double the weighting of the 1_{st} bit.

Now lets see what happens to the output voltage if we connect input bit V_C HIGH to +5 volts.

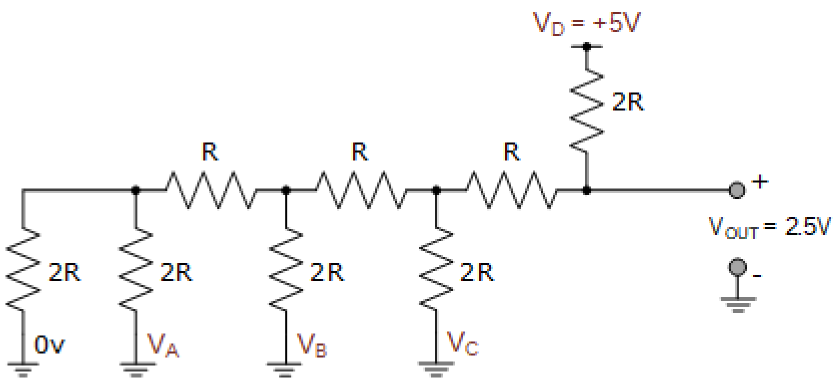
R-2R DAC with Input V_C



With input V_C HIGH and logic level "1" and the other input bits at logic level "0", the output voltage, V_{OUT} is calculated at 1.25 volts, and which is one-quarter (1/4) the value of the +5V input (5/1.25 = 4) voltage. Again we can see that this voltage is double the output of input bit V_B but also 4 times the value of bit V_A. This is because input V_C is the 3_{rd} bit so has double the weighting of the 2_{nd} bit and four times the weighting of the 1_{st} bit.

Finally lets see what happens to the output voltage if we connect input V_D HIGH to +5 volts.

R-2R DAC with Input V_D



With only input V_D HIGH and logic level "1" and the other inputs at logic level "0", the output voltage, V_{OUT} is calculated at 2.5 volts. This is on-half (1/2) the value of the +5V input (5/2.5 = 2) voltage. Again we can see that this voltage is double the output of input bit V_C, 4 times the value of bit V_B and 8 times the value of input bit V_A as it is the 4_{th} bit and therefore classed as the Most Significant Bit, (MSB).

Then we can see that if input V_A represents the LSB and therefore controls the DAC's resolution, and input V_B is double V_A, input V_C is four times greater than V_A, and input V_D is eight times greater than V_A, we can obtain a relationship for the analogue output voltage of our 4-bit digital-to-analogue converter with the following equation:

Digital-to-Analogue Output Voltage Equation

$$V_{OUT} = \frac{V_A + 2V_B + 4V_C + 8V_D}{16}$$

Where the denominator value of 16 corresponds to the 16 (2^4) possible combinations of inputs to the 4-bit R-2R ladder network of the DAC.

We can expand this equation further to obtain a generalised R-2R DAC equation for any number of digital inputs for a R-2R D/A converter as the weighting of each input bit will always be referenced to the least significant bit (LSB), giving us a generalised equation of:

Generalised R-2R DAC Equation

$$V_{OUT} = \frac{V_A + 2V_B + 4V_C + 8V_D + 16V_E + 32V_F + \ldots \text{etc}}{2^n}$$

Where: "n" represents the number of digital inputs within the R-2R resistive ladder network of the DAC producing a resolution of: $V_{LSB} = V_{IN}/2^n$.

Clearly then input bit V_A when HIGH will cause the smallest change in the output voltage, while input bit V_D when HIGH will cause the greatest change in the output voltage. The expected output voltage is therefore calculated by summing the effect of all the individual input bits which are connected HIGH.

Ideally, the ladder network should produce a linear relationship between the input voltages and the analogue output as each input will have a step increase equal to the LSB, we can create a table of expected output voltage values for all 16 combinations of the 4 inputs with +5V representing a logic "1" condition as shown.

4-bit R-2R D/A Converter Voltage Output

Digital Inputs				V_{OUT} Expression	V_{OUT}
D	C	B	A	$(8*V_D + 4*V_C + 2*V_B + 1*V_A)/2^4$	in Volts
0	0	0	0	(0*5 + 0*5 + 0*5 + 0*5)/16	0
0	0	0	1	(0*5 + 0*5 + 0*5 + 1*5)/16	0.3125
0	0	1	0	(0*5 + 0*5 + 2*5 + 0*5)/16	0.6250
0	0	1	1	(0*5 + 0*5 + 2*5 + 1*5)/16	0.9375
0	1	0	0	(0*5 + 4*5 + 0*5 + 0*5)/16	1.2500
0	1	0	1	(0*5 + 4*5 + 0*5 + 1*5)/16	1.5625
0	1	1	0	(0*5 + 4*5 + 2*5 + 0*5)/16	1.8750
0	1	1	1	(0*5 + 4*5 + 2*5 + 1*5)/16	2.1875
1	0	0	0	(8*5 + 0*5 + 0*5 + 0*5)/16	2.5000
1	0	0	1	(8*5 + 0*5 + 0*5 + 1*5)/16	2.8125

1	0	1	0	(8*5 + 0*5 + 2*5 + 0*5)/16	3.1250
1	0	1	1	(8*5 + 0*5 + 2*5 + 1*5)/16	3.4375
1	1	0	0	(8*5 + 4*5 + 0*5 + 0*5)/16	3.7500
1	1	0	1	(8*5 + 4*5 + 0*5 + 1*5)/16	4.0625
1	1	1	0	(8*5 + 4*5 + 2*5 + 0*5)/16	4.3750
1	1	1	1	(8*5 + 4*5 + 2*5 + 1*5)/16	4.6875

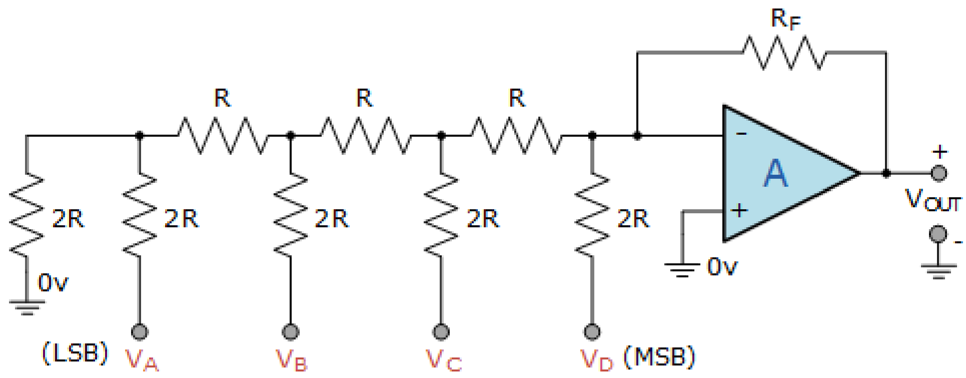
You may have noticed that the full-scale analogue output voltage for a binary code of **1111** never reaches the same value as the digital input voltage (+5V) as it is always less by the equivalent of one LSB bit, (312.5mV in this example).

However, the higher the number of digital input bits (resolution) the nearer the analogue output voltage reaches full-scale when all the input bits are HIGH. Likewise when all the input bits are LOW, the resulting lower resolution of LSB makes V_{OUT} closer to zero volts.

R-2R Digital-to-Analogue Converter

Now that we understand what a *R-2R resistive ladder network* is and how it works, we can use it to produce a R-2R Digital-to-Analogue Converter. Again using our 4-bit R-2R resistive ladder network from above and adding it to an inverting operational amplifier circuit, we can create a simple R-2R digital-to-analogue converter of:

R-2R Digital-to-Analogue Converter



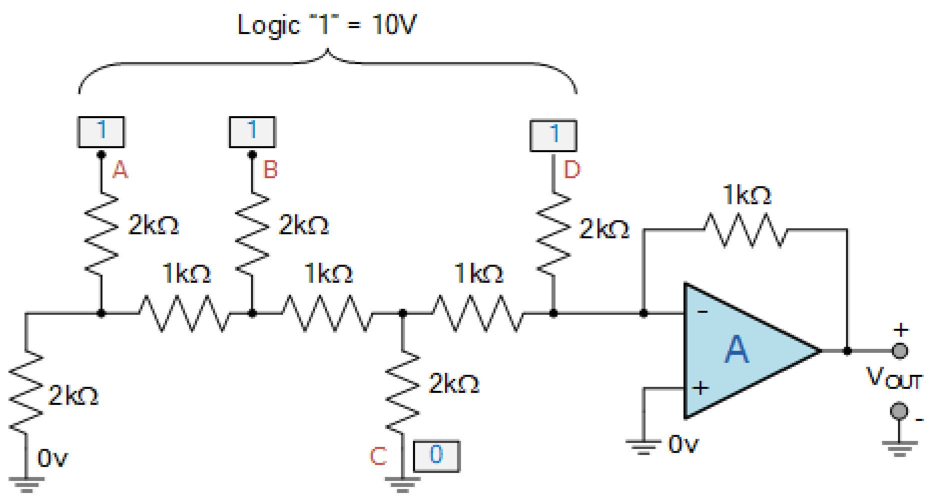
The digital logic circuit used to drive the D/A converter can be generated by combinational or sequential logic circuits, data registers, counters or simply switches. The interfacing of a R-2R D/A converter of “n”-bits will depend upon its application. All-in-one boards such as the Arduino or Raspberry Pi have *digital-to-analogue converters* built-in so make interfacing and programming much easier. There are many popular DAC’s available such as the 8-bit DAC0808.

R-2R D/A Converter Example No1

A 4-bit R-2R digital-to-analogue converter is constructed to control the speed of a small DC motor using the output from a digital logic circuit. If the logic circuit uses 10 volt CMOS devices, calculate the analogue output voltage from the DAC when the input code is hexadecimal number “B”. Also determine the resolution of the DAC.

1). The hexadecimal letter “B” is equal to the number eleven in decimal. The decimal number eleven is equal to the binary code “1011” in binary. That is: $B_{16} = 1011_2$. Thus for our 4-bit binary number of 1011_2, input bit D = 1, bit C = 0, bit B = 1 and bit A = 1.

If we assume that feedback resistor R_F is equal to “R”, then our R-2R D/A converter circuit will look like:



The digital logic circuit uses 10 volt CMOS devices, so the input voltage to the R-2R network will be 10 volts. Also being a 4-bit ladder DAC, there will be 2^4 possible input combinations, so using our equation from above, the ouput voltage for a binary code of 1011_2 is calculated as:

$$V_{OUT} = \frac{1V_A + 2V_B + 4V_C + 8V_D}{2^n}$$

$$V_{OUT} = \frac{1\times10 + 2\times10 + 4\times0 + 8\times10}{16}$$

$$V_{OUT} = \frac{110}{16} = 6.875 \text{ Volts}$$

Therefore the analogue output voltage used to control the DC motor when the input code is 1011_2 is calculated as: -6.875 volts. Note that the output voltage is negative due to the inverting input of the operational amplifier.

2). The resolution of the converter will be equal to the value of the least significant bit (LSB) which is given as:

$$\text{Resolution} = V_{(LSB)} = \frac{V_{IN}}{2^n}$$

$$\therefore \ \text{Resolution} = \frac{10}{16} = 0.625 \text{ Volts}$$

Then the smallest step change of the analogue output voltage, V_{OUT} for a 1-bit LSB change of the digital input of this 4-bit R-2R digital-to-analogue converter example is: 0.625 volts. That is the output voltage changes in steps or increments of 0.625 volts and not as a straight linear value.

4-bit Binary Counting R-2R DAC

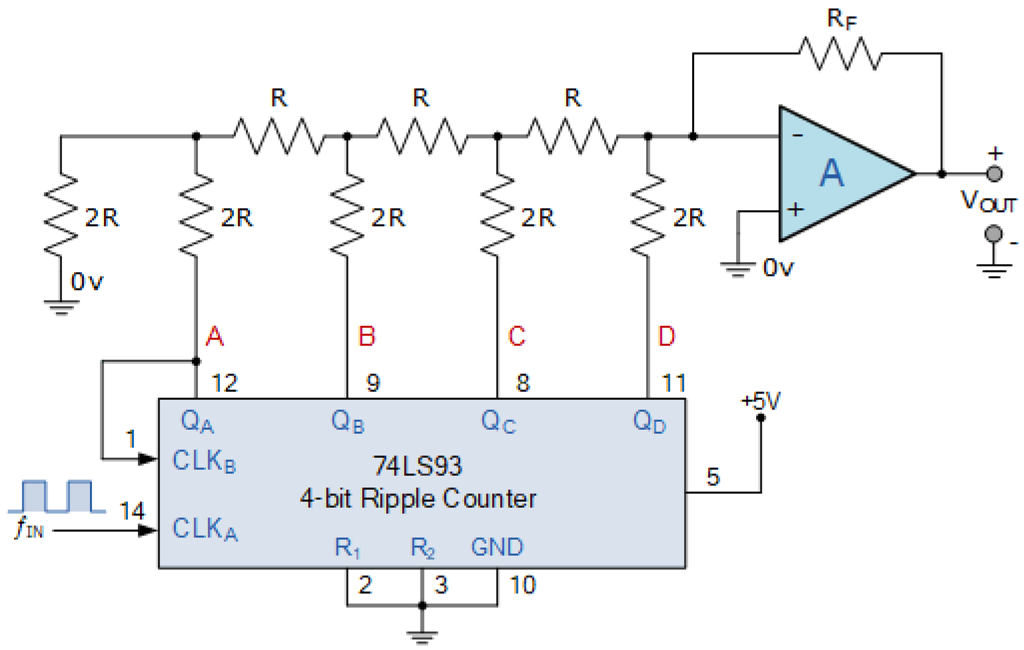
Hopefully by now we understand that we can make a R-2R ladder DAC using just two resistor values, one the base value "R" and the other twice or double the value being "2R". In our simple example above we have made a 4-bit R-2R DAC with four input data lines, A, B, C, and D giving us 16 (2^4) different input combinations from "0000" to "1111".

The binary code for these four digital input lines can be generated in many different ways, using micro-controllers, digital circuits, mechanical or solid state switches. But one interesting option is to use a 4-bit binary counter such as the 74LS93.

The **74LS93** is a 4-bit J-K ripple counter which can be configured to count-up from 0000_2 to 1111_2 (MOD-16) and reset back to zero (0000) again by the application of a single external clock signal. The 74LS93 is an asynchronous counter commonly called a "ripple" counter because of the way that the internal J-K bistables respond to the clock or timing input producing a 4-bit binary output.

The frequency (or period) of this external clock or timing pulse is divided by a factor of 2, 4, 8, and 16 by the counters output lines as the clock pulse appears to ripple through the four J-K flip-flops producing the required 4-bit output count sequence from 0000_2 to 1111_2.

4-bit Binary Counting R-2R DAC



Note that to count upwards from 0000 to 1111, the external CLK_B input must be connected to the Q_A (pin-12) output and the input timing pulses are applied to input CLK_A (pin-14).

This simple 4-bit asynchronous up counter built around the 74LS93 binary ripple counter as the same counting sequence given in the above table. On the application of a clock pulse the outputs: Q_A, Q_B, Q_C, and Q_D change by one step.

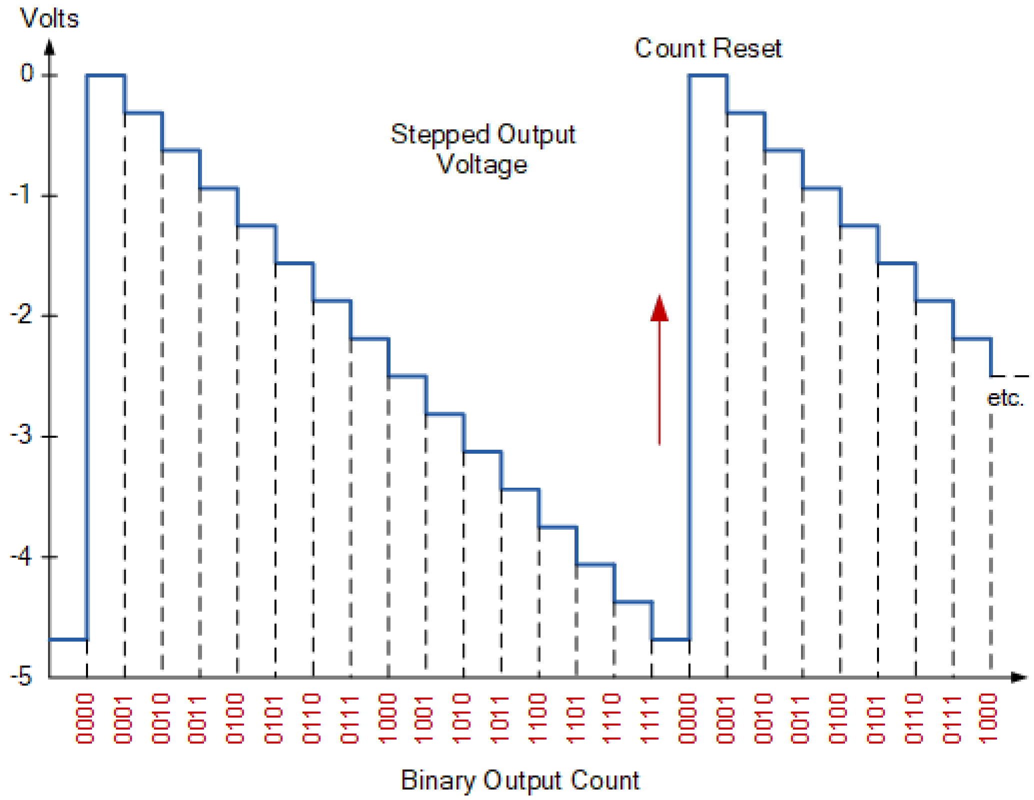
The input of the operational amplifier detects this step change and outputs a negative voltage (inverting op-amp) relative to the binary code at the R-2R ladder inputs. The output voltage value for each step will correspond to that given in the table above.

The ripple counter will count up in sequence with the four outputs producing an output sequence of binary values upto the 15th clock pulse where the outputs are set to 1111_2 (decimal 15) producing the maximum negative output voltage of the digital-to-analogue converter.

On the 16th pulse the counters output sequence is reset and the count returns back to 0000, which resets the op-amps output back to zero volts. The application of the next clock pulse begins a new counting cycle from zero to $V_{OUT(max)}$.

We can show the output sequence for this simple 4-bit binary asynchronous counting R-2R D/A converter in the following timing diagram.

4-bit R-2R DAC Timing Diagram

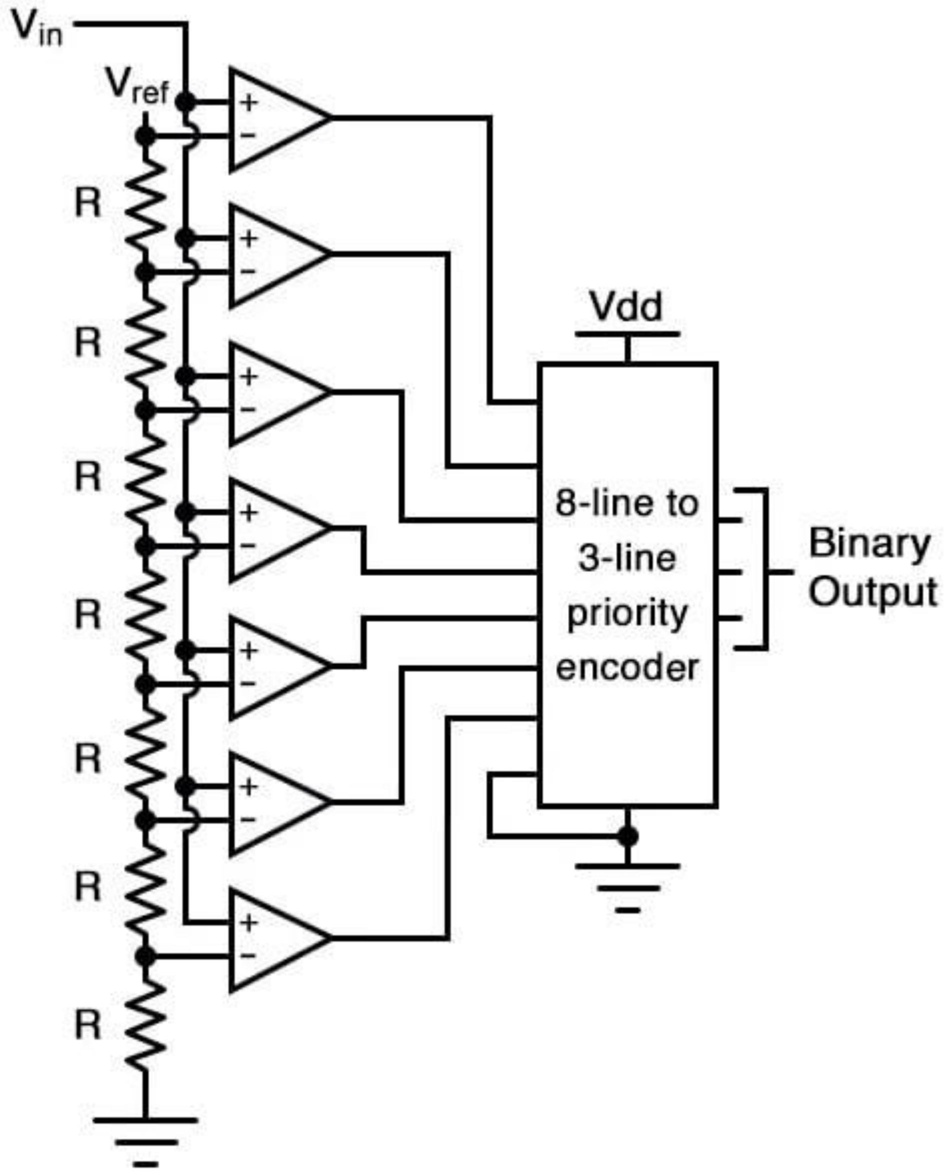


Clearly then, the output voltage of the operational amplifier varies from zero volts to its maximum negative voltage as the ripple counter counts from 0000_2 to 1111_2 respectively. This simple circuit could be used to vary the brightness of a lamp connected to the op-amps output, or continually vary the speed of a DC motor from slow to fast, and back to slow again at a rate determined by the clock period.

Here the ripple counter and R-2R DAC are configured for 4-bit operation but using commonly available binary ripple counters such as the CMOS 4024 7-bit (÷128), the CMOS 4040 12-bit (÷4096) or the larger CMOS 4060 14-bit (÷16,384) counter and adding more input resistors to the R-2R ladder network such as those available from **Bournes**, the resolution (LSB) of the circuit can be greatly lowered producing a smoother output signal from the *R-2R digital-to-analogue converter*.

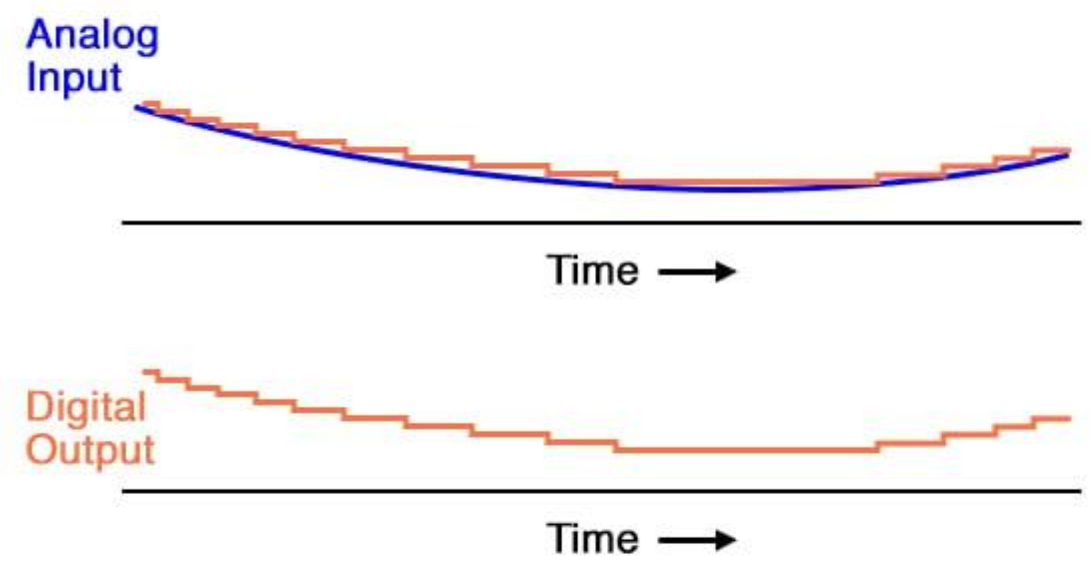
Flash ADC

Also called the *parallel* A/D converter, this circuit is the simplest to understand. It is formed of a series of comparators, each one comparing the input signal to a unique reference voltage. The comparator outputs connect to the inputs of a priority encoder circuit, which then produces a binary output. The following illustration shows a 3-bit flash ADC circuit:

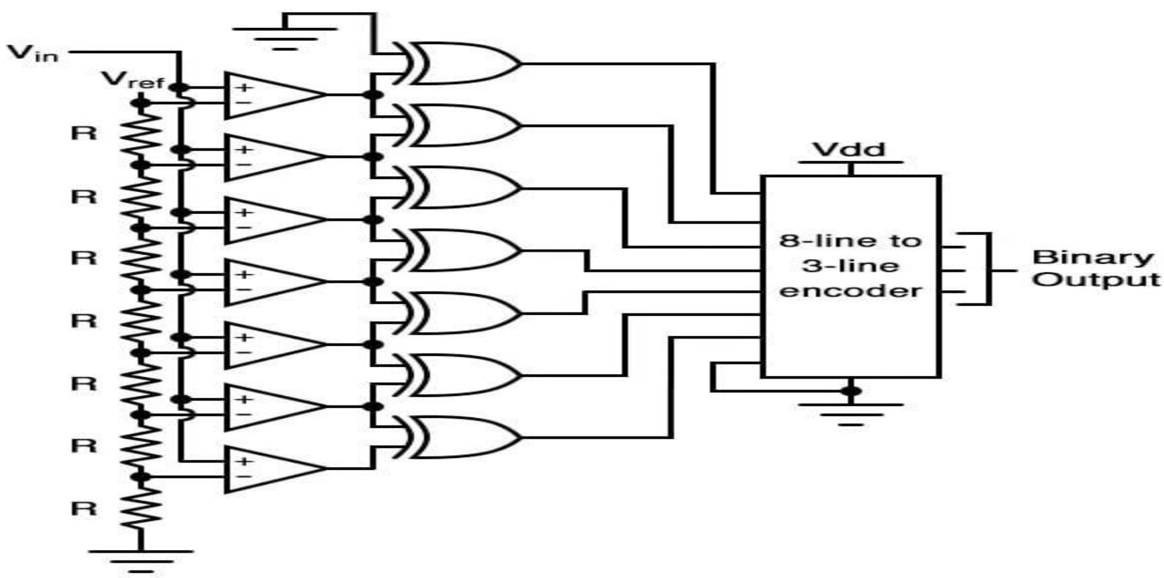


V_{ref} is a stable reference voltage provided by a precision voltage regulator as part of the converter circuit, not shown in the schematic. As the analog input voltage exceeds the reference voltage at each comparator, the comparator outputs will sequentially saturate to a high state. The priority encoder generates a binary number based on the highest-order active input, ignoring all other active inputs.

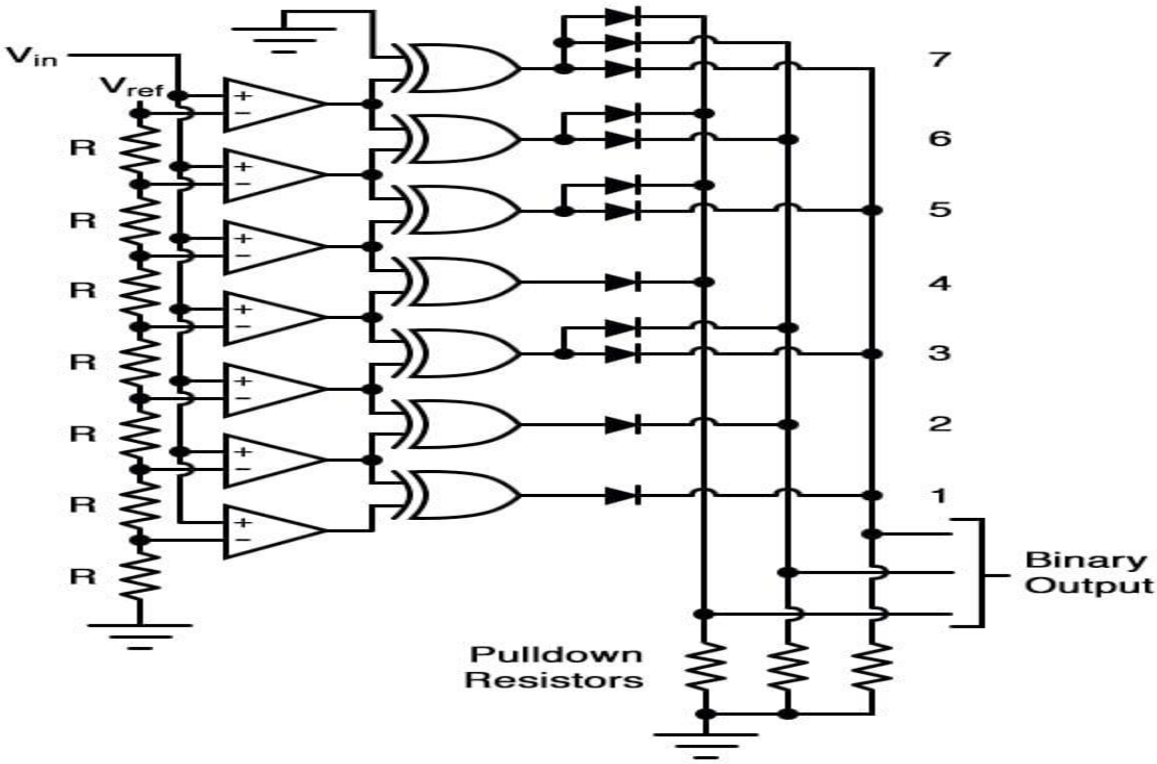
When operated, the flash ADC produces an output that looks something like this:



For this particular application, a regular priority encoder with all its inherent complexity isn't necessary. Due to the nature of the sequential comparator output states (each comparator saturating "high" in sequence from lowest to highest), the same "highest-order-input selection" effect may be realized through a set of Exclusive-OR gates, allowing the use of a simpler, non-priority encoder:



And, of course, the encoder circuit itself can be made from a matrix of diodes, demonstrating just how simply this converter design may be constructed:



Not only is the flash converter the simplest in terms of operational theory, but it is the most efficient of the ADC technologies in terms of speed, being limited only in comparator and gate propagation delays. Unfortunately, it is the most component-intensive for any given number of output bits.

This three-bit flash ADC requires seven comparators. A four-bit version would require 15 comparators. With each additional output bit, the number of required comparators doubles.

Considering that eight bits is generally considered the minimum necessary for any practical ADC (255 comparators needed!), the flash methodology quickly shows its weakness. An additional advantage of the flash converter, often overlooked, is the ability for it to produce a non-linear output.

With equal-value resistors in the reference voltage divider network, each successive binary count represents the same amount of analog signal increase, providing a proportional response. For special applications, however, the resistor values in the divider network may be made non-equal.

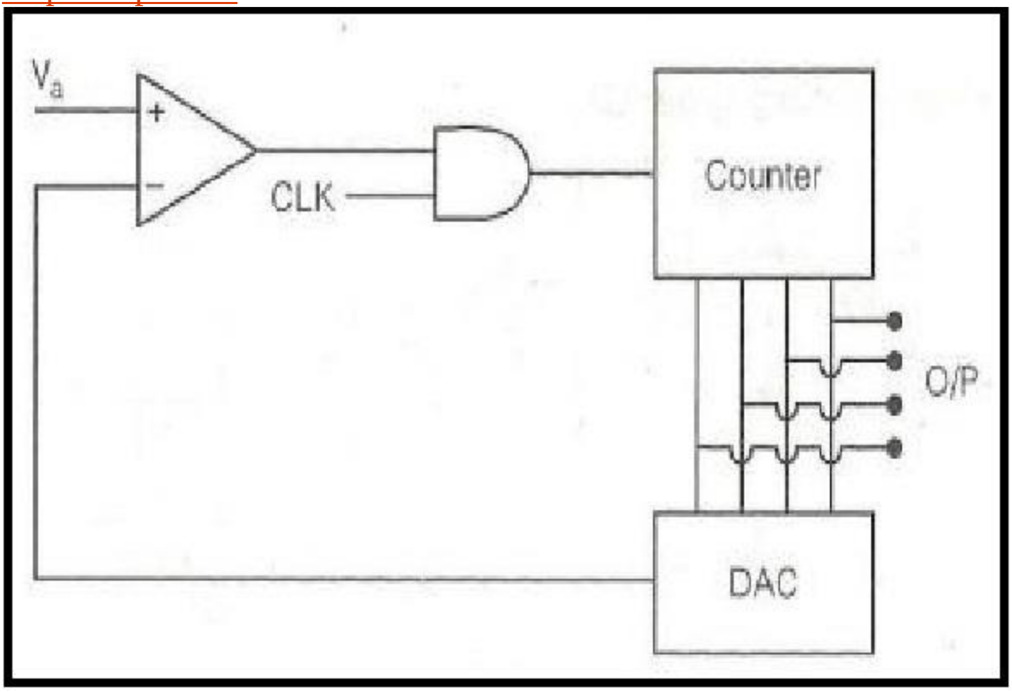
This gives the ADC a custom, nonlinear response to the analog input signal. No other ADC design is able to grant this signal-conditioning behavior with just a few component value changes.

Designing of Counter Type ADC

In electronics, the term "analog to digital conversion" can be denoted with ADC, A/D or A to D. It is one kind of system used to convert analog signal to digital signal. An A/D may also give an inaccessible dimension like an electronic device that changes an analog i/p current or voltage to a digital number representing the magnitude of the voltage or current. Normally the digital o/p is a 2's complement binary number that is relative to the i/p, but there are other possibilities. There are numerous ADC architectures, but some of the specific ADCs ate implemented as ICs (integrated circuits) due to the complexity and the requirement for exactly matched components. A digital-to-analog converter (DAC) performs the reverse function; it converts a digital signal into an analog signal. The different types of ADCs available in different speeds, interfaces and accuracy, namely a Flash type ADC, Counter type ADC, sigma-delta ADC and successive approximation ADC.

What is Counter Type ADC?

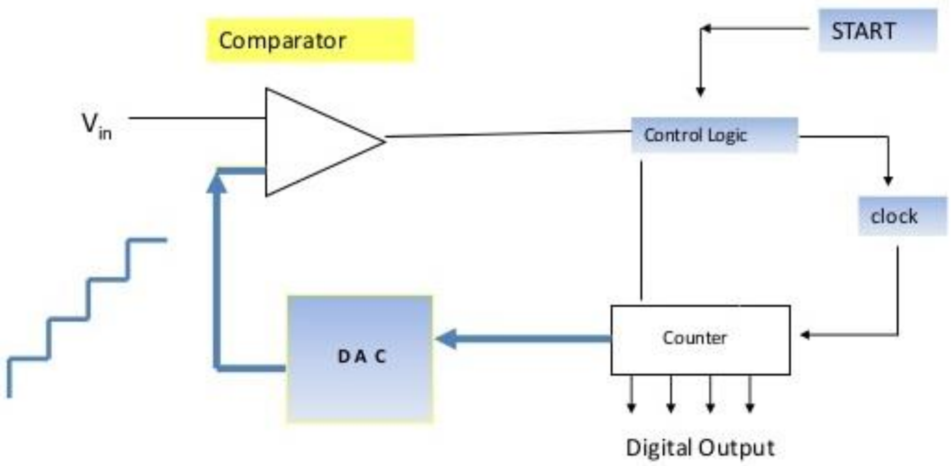
The Counter type ADC can be defined as, it is the basic type of ADC, which is also known as staircase approximation ADC, or a ramp type ADC. The circuit diagram of counter type ADC is shown below. The circuit diagram of Counter Type ADC can be built with N-bit counter, digital to analog converter, and op-amp comparator.



Counter Type ADC

Counter Type ADC Operation

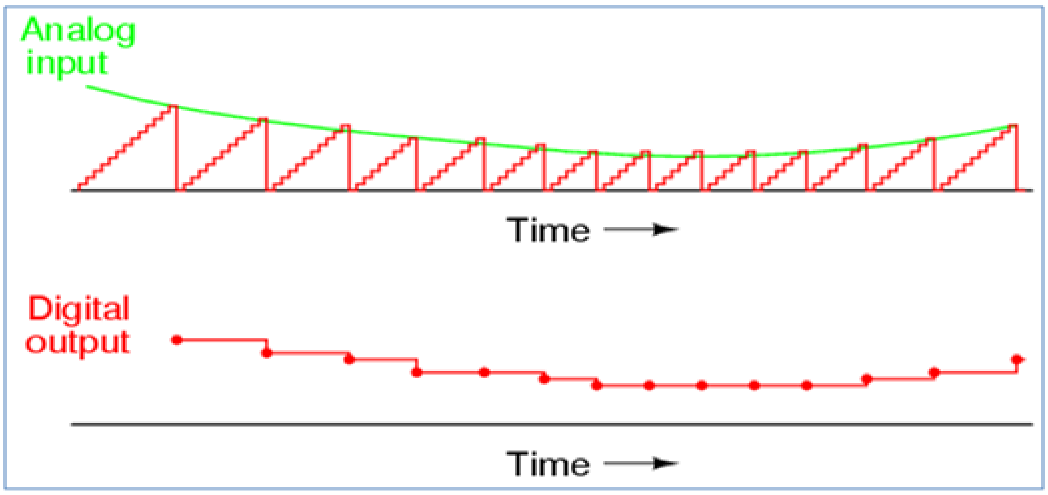
The N-bit counter produces an n-bit digital o/p which is given as an i/p to the digital to analog circuit (DAC). The analog output equivalent to the digital i/p from DAC is contrasted with the i/p analog voltage with the help of an op-amp comparator. This Integrated Circuit evaluates the two voltages and if the produced DAC voltage is low, it gives a high pulse to the N-bit counter as a CLK pulse to raise the counter.



Counter Type ADC Operation

The similar procedure will be continued until the output of the DAC equals to the i/p analog voltage then it produces a low CLK pulse and also gives a clear signal to the counter as well as a load signal to the storage resistor. Here storage resistor is used to store the corresponding digital bits. These digital values are strongly matched with the analog input values with a small error.

For each sampling interval, the output of DAC tracks a rampway so that it is named as a Digital ramp kind ADC. And this ramp seems like staircases for each sampling moment, so that it is also named as a staircase approximation kind ADC.



Counter Type ADC Waveforms

Counter Type ADC Conversion Time

The ADC conversion time is the time taken the process to change the input sampled analog price to a digital value. Here the most conversions of high i/p voltage for an N-bit ADC is the CLK pulses necessary to the counter to calculate its maximum count value. So

The Counter type ADC conversion can be done by this formula, that is = (2N-1) T

Where ‘T’ is the time period of the CLK pulse.

If N=3 bits, then the Tmax = 7T.

By viewing the above change time of Counter type ADC it is demonstrated that the sampling phase of Counter type ADC should be as shown below.

Ts >=(2N-1) T

Counter type ADC Advantages

- Counter type ADC is very simple to understand and also to operate.
- Counter type ADC design is less complex, so the cost is also less

Counter type ADC Disadvantages

- Speed is less, since each time the counter has to begin from ZERO.
- There may be conflicts if the next i/p is sampled before completion of one process.

Successive Approximation type ADC

Successive Approximation type ADC is the most widely used and popular ADC method. The conversion time is maintained constant in successive approximation type ADC, and is proportional to the number of bits in the digitaloutput, unlike the counter and continuous type A/D converters. The basic principle of this type of A/D converter is that the unknown analog input voltage is approximated against an n-bit digital value by trying one bit at a time, beginning with the MSB. The principle of successive approximation process for a 4-bit conversion is explained here. This type of ADC operates by successively dividing the voltage range by half, as explained in the following steps.

(1) The MSB is initially set to 1 with the remaining three bits set as 000. The digital equivalent voltage is compared with the unknown analog input voltage.

(2) If the analog input voltage is higher than the digital equivalent voltage, the MSB is retained as 1 and the second MSB is set to 1. Otherwise, the MSB is set to 0 and the second MSB is set to 1. Comparison is made as given in step (1) to decide whether to retain or reset the second MSB.

The above steps are more accurately illustrated with the help of an example.

Let us assume that the 4-bit ADC is used and the analog input voltage is VA = 11 V. when the conversion starts, the MSB bit is set to 1.

Now VA = 11V > VD = 8V = [1000]2

Since the unknown analog input voltage VA is higher than the equivalent digital voltage VD, as discussed in step (2), the MSB is retained as 1 and the next MSB bit is set to 1 as follows

VD = 12V = [1100]2

Now VA = 11V < VD = 12V = [1100]2

Here now, the unknown analog input voltage VA is lower than the equivalent digital voltage VD. As discussed in step (2), the second MSB is set to 0 and next MSB set to 1 as

VD = 10V = [1010]2

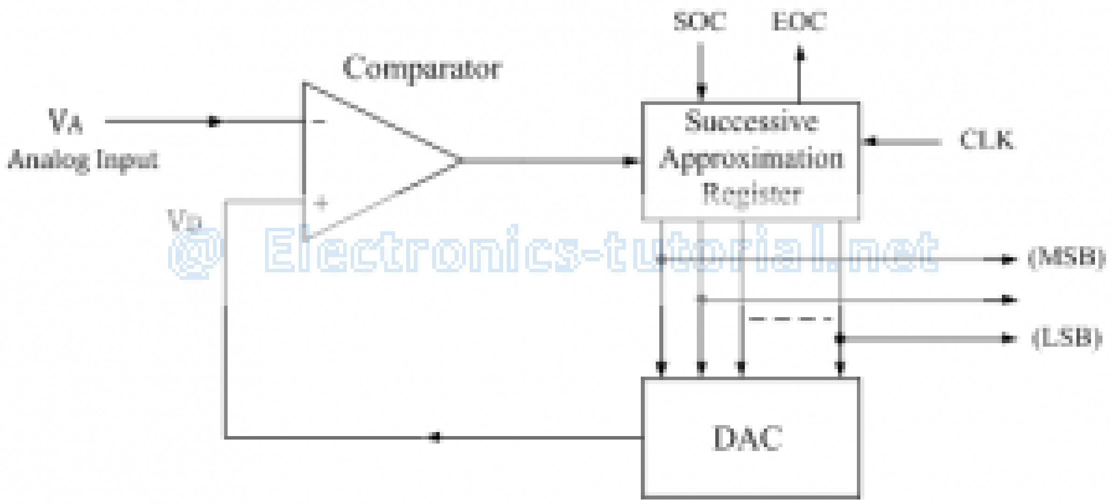
Now again VA = 11V > VD = 10V = [1010]2

Again as discussed in step (2) VA>VD, hence the third MSB is retained to 1 and the last bit is set to 1. The new code word is

VD = 11V = [1011]2

Now finally VA = VD , and the conversion stops.

The functional block diagram of successive approximation type of ADC is shown below.



It consists of a successive approximation register (SAR), DAC and comparator. The output of SAR is given to n-bit DAC. The equivalent analog output voltage of DAC, VD is applied to the non-inverting input of the comparator. The second input to the comparator is the unknown analog input voltage VA. The output of the comparator is used to activate the successive approximation logic of SAR.
When the start command is applied, the SAR sets the MSB to logic 1 and other bits are made logic 0, so that the trial code becomes 1000.

Advantages:
1 Conversion time is very small.
2 Conversion time is constant and independent of the amplitude of the analog input signal VA.

Disadvantages:
1 Circuit is complex.
2 The conversion time is more compared to flash type ADC.